AMERICAN INTERNATIONAL UNIVERSITY - BANGLADESH (AIUB)

From Resolution to Explanation: Real-ESRGAN and LIME Analysis of Vision Transformers and CNNs for Brain Tumor MRI Classification

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Introduction

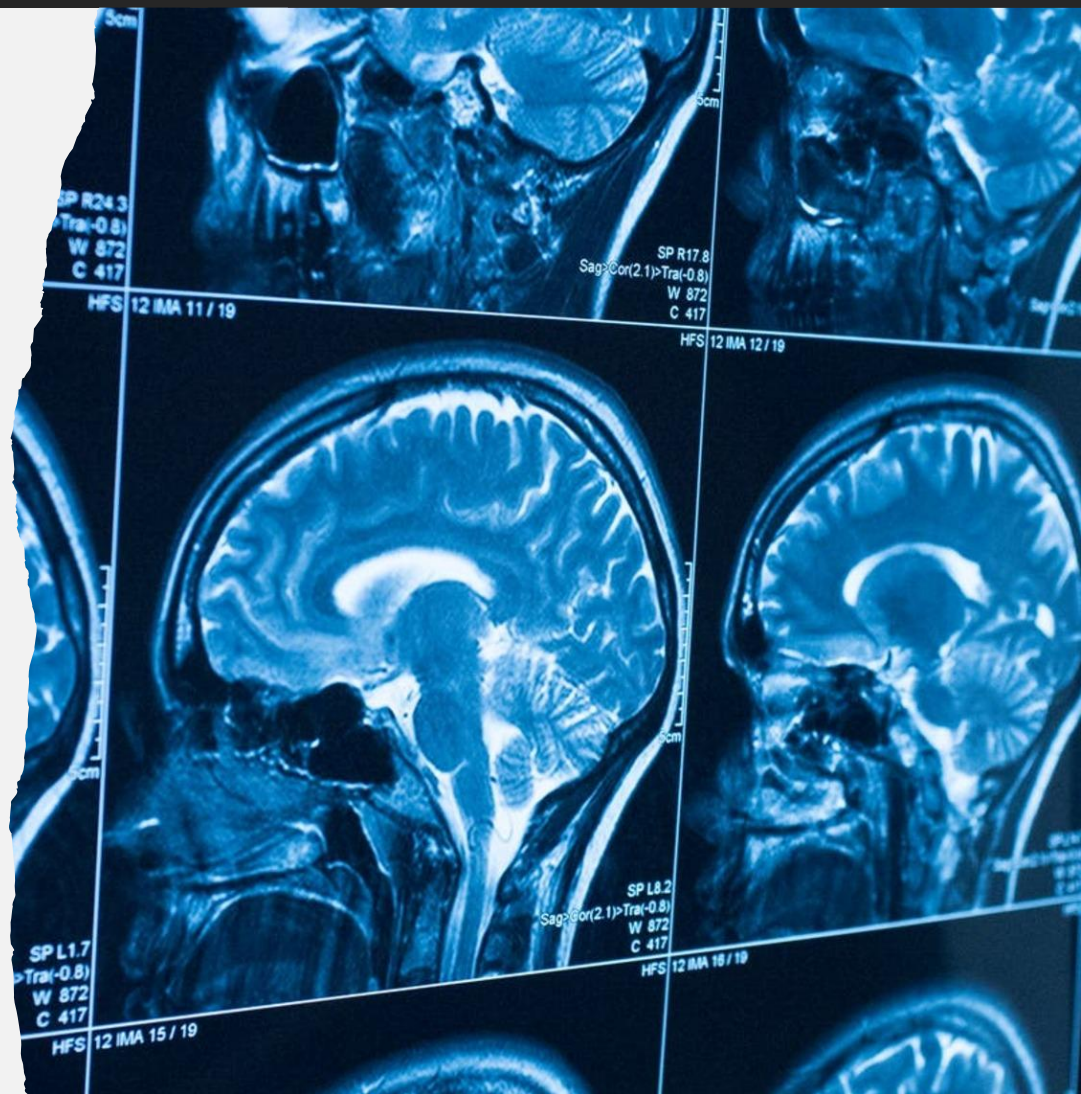
Brain tumors (e.g., glioma, meningioma, pituitary) pose diagnostic challenges due to irregular boundaries and morphology.

Deep learning (DL) is key for accurate MRI classification; timely detection improves outcomes.

CNNs: Strong in local features (e.g., ResNet50 >99% accuracy).

ViTs: Excel in global context via self-attention.

Our Contribution: Benchmark ViTs vs. CNNs with Real-ESRGAN (Real-Enhanced Super Resolution Generative Adversarial Networks) enhancement and LIME for interpretability.



Background Study

Real-ESRGAN: Advanced super-resolution GAN; uses RRDBs for real-world degradations (noise, blur); enhances MRI textures without distorting anatomy.

LIME: Post-hoc XAI; perturbs inputs (e.g., superpixels) to approximate local model behavior; highlights influential regions.

Performance Metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix (equations from paper).

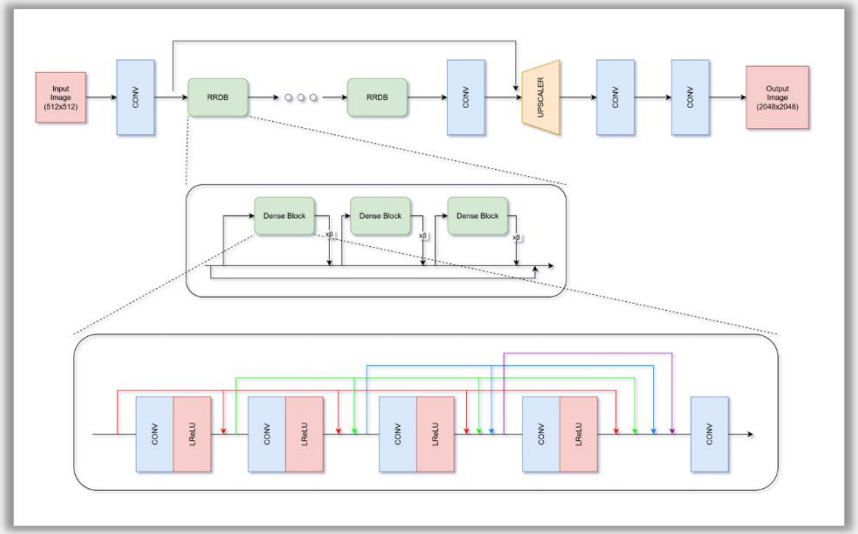


Fig. 1: Real-ESRGAN Architecture

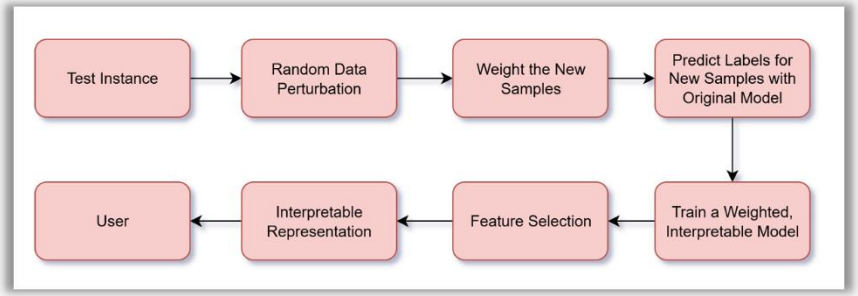


Fig. 2: LIME Architecture

Existing Work

- Krishnan et al. (2024): Rotation Invariant Vision Transformer (RViT) embeds rotating patches for orientation robustness, achieving 98.6% accuracy on Kaggle dataset. **[1]**

- Ahmed et al. (2024): Hybrid ViT–GRU with XAI (LIME, SHAP, attention maps) on BrTMHD-2023, reaching 98.97% accuracy and 97% F1-score for interpretable detection. **[2]**

- Zeineldin et al. (2024): TransXAI (CNN–ViT hybrid) for multimodal glioma segmentation, generating clinically interpretable heatmaps. **[3]**

- Sahu (2025): Compared ViT-B16 vs. EfficientNetB0; ViT at 71.6% accuracy with better generalization and Grad-CAM localization in low-data scenarios. **[4]**

- Kawadkar (2025): DeiT-Small outperformed ResNet-50 and EfficientNet-B0 at 92.16% accuracy in brain tumor classification benchmarks. **[5]**

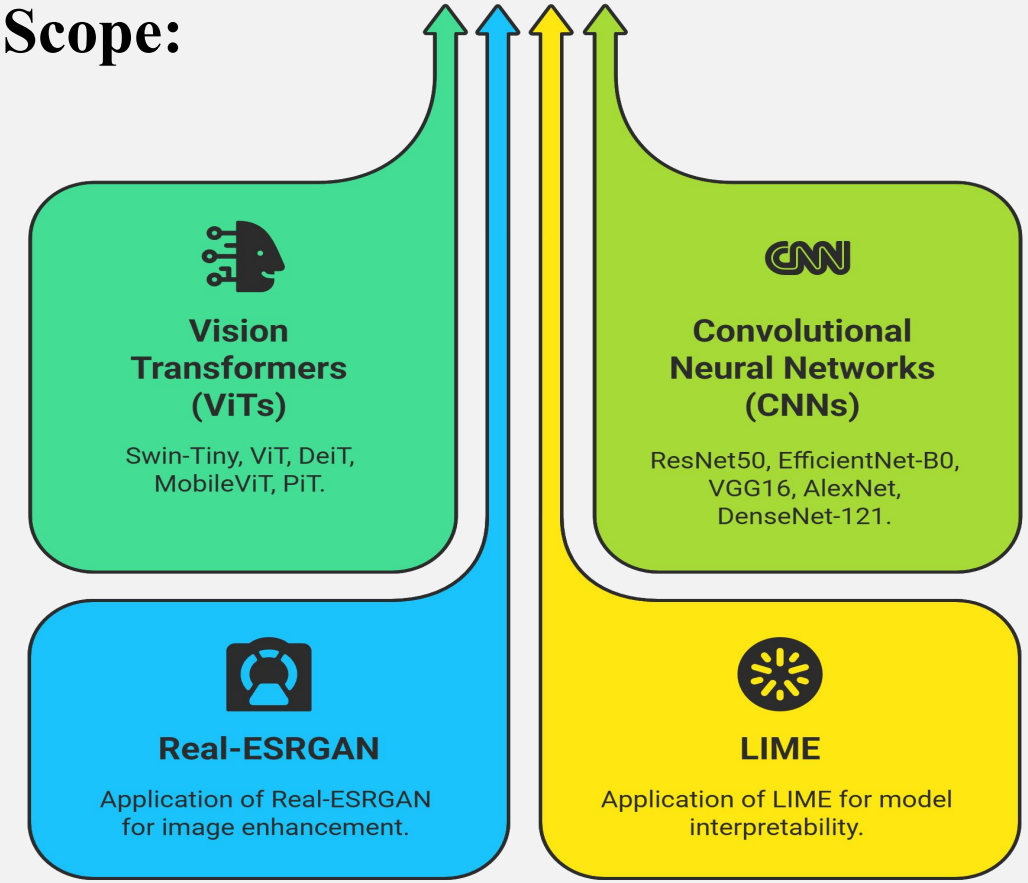
- Karagoz et al. (2024): Self-supervised ResViT pre-trained on MRI synthesis, achieving 98.5% accuracy in data-scarce

Research Scope & Objective

Objective:

- Enhance MRI quality for better classification.
- Compare ViT global vs. CNN local focus.
- Provide interpretable insights for clinical use.
- Propose hybrid ViT-ResNet50 for robustness.

Scope:



Made with Napkin

Dataset

Source: Kaggle Brain Tumor MRI (~7,000 images).

Classes: Glioma, Meningioma, Pituitary, No Tumor.

Examples: Varied tumor shapes/sizes (Fig. 4).

Distribution: ~1,750 images/class (Fig. 3).

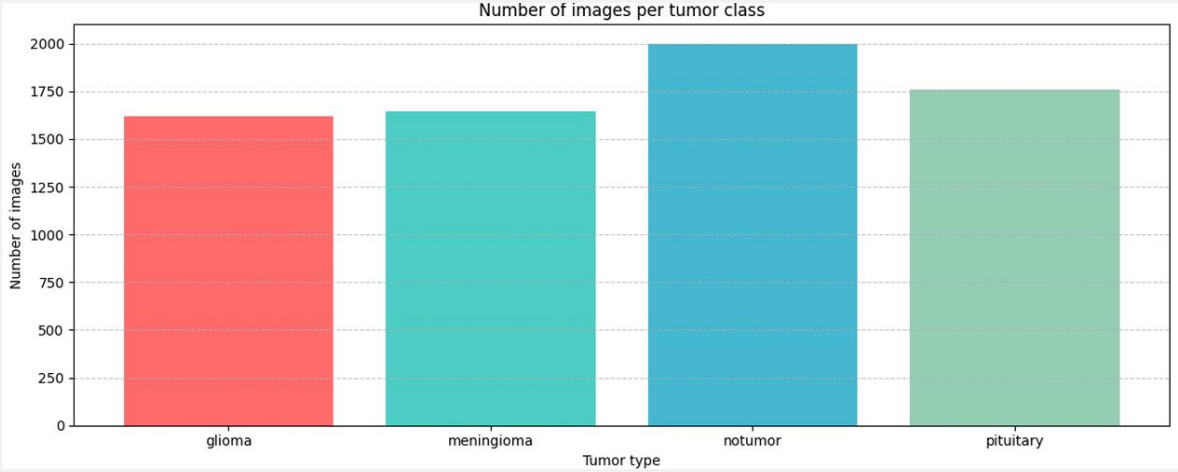


Fig. 3: Image per tumor class

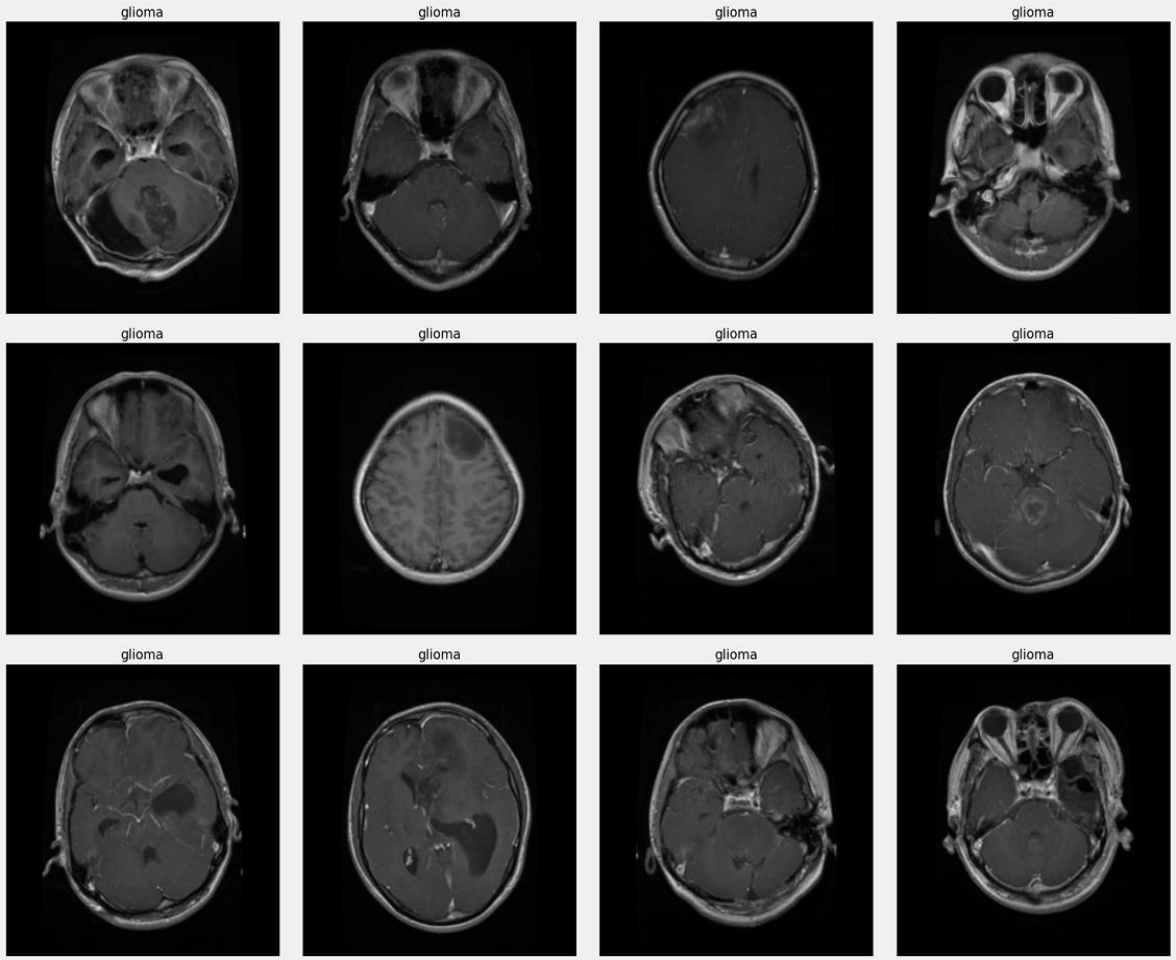


Fig. 4: Some images from the dataset.

Research Methodology

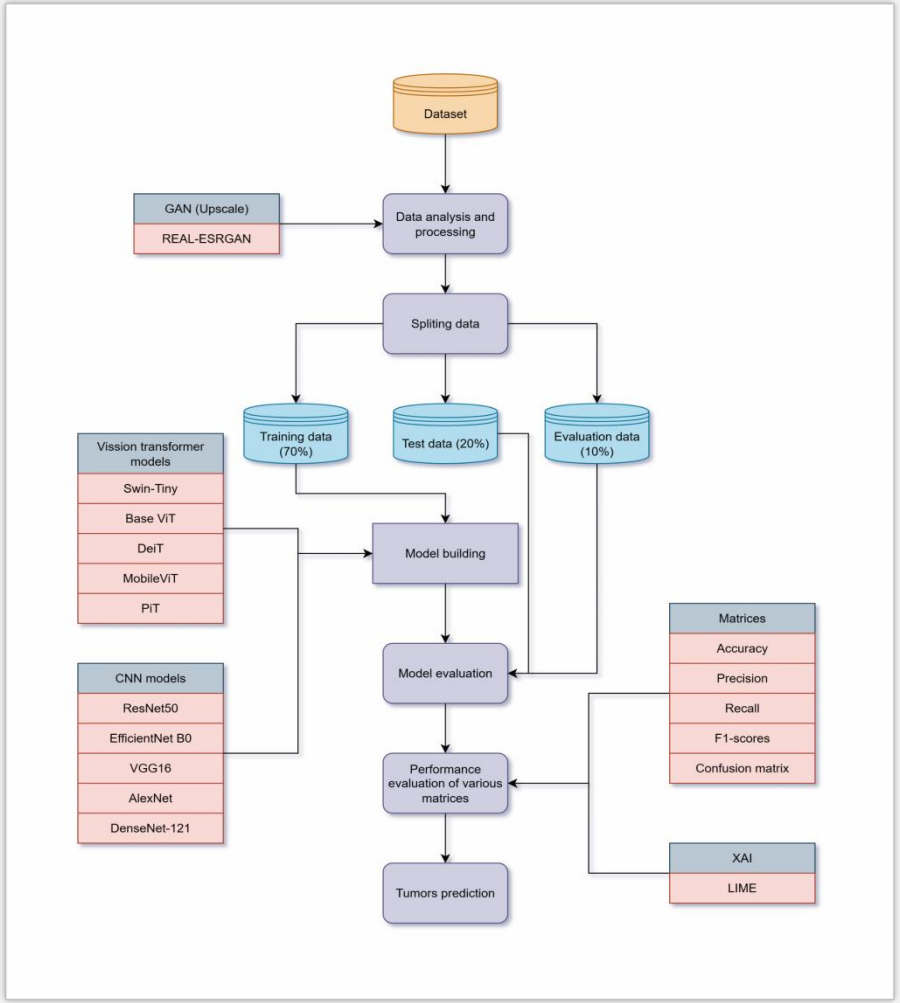


Fig. 5: Proposed methodology for research work

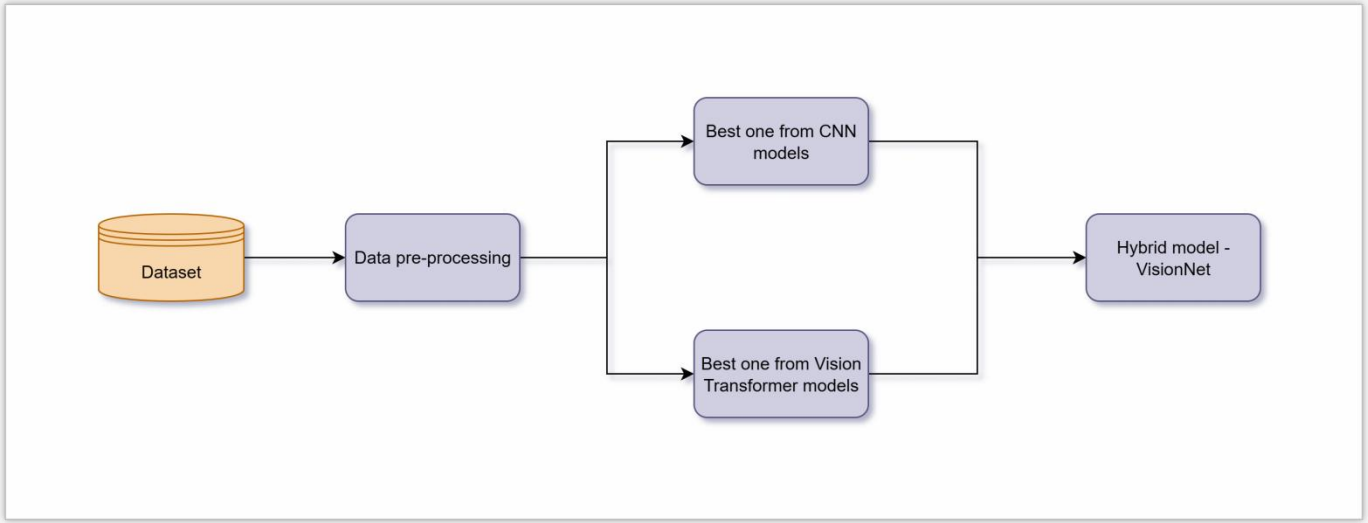
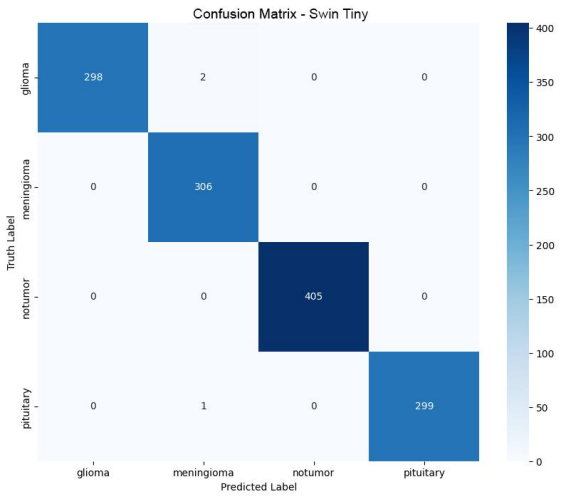
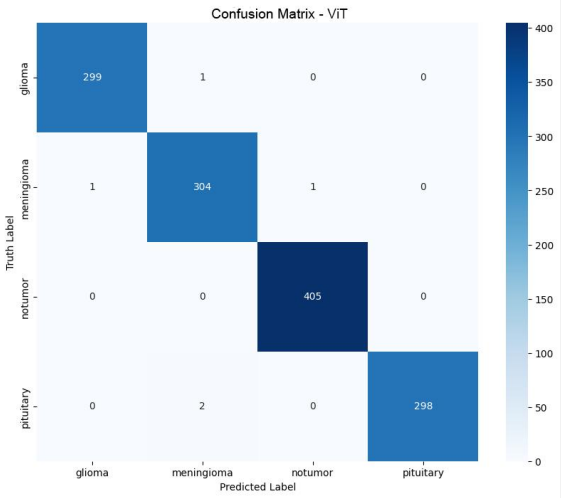


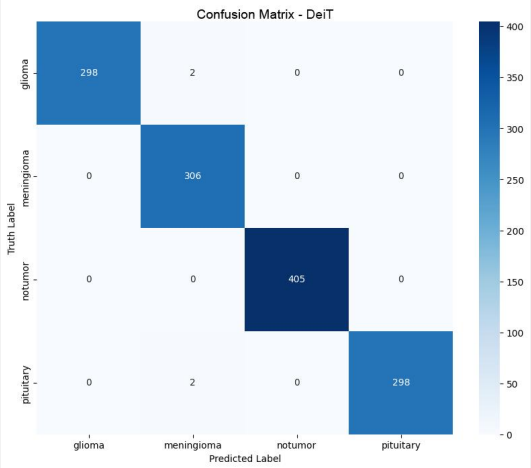
Fig. 6: Proposed hybrid model VisionNet

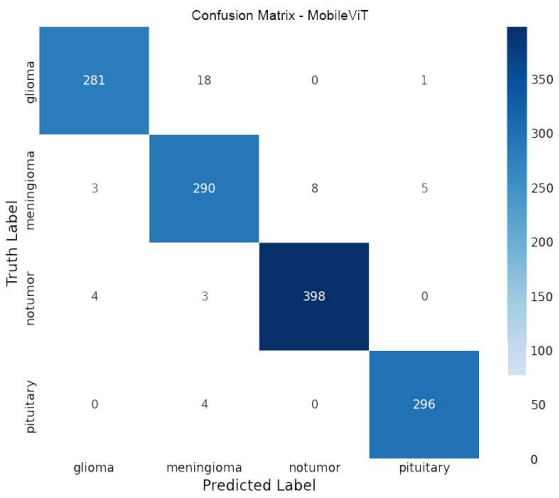
ViT Models Performance Analysis

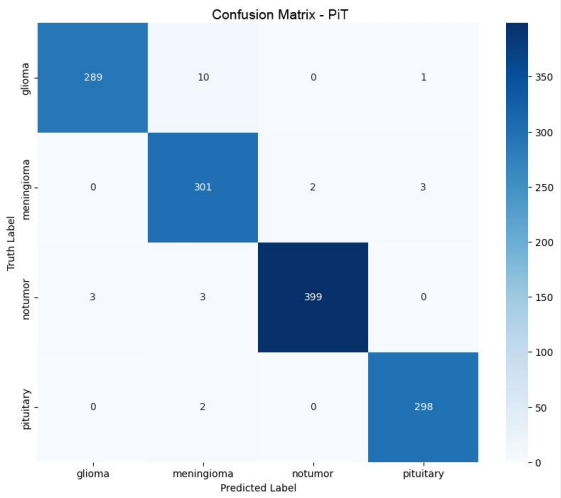
Model Name	Accuracy	Precision	Recall	F1-Score
Swin-Tiny	99.76%	99.75%	99.75%	100%
ViT	99.77%	99.75%	99.75%	100%
DeiT	99.69%	99.75%	99.50%	99.75%
MobileViT	96.49%	96.50%	96.50%	96.25%
PiT	98.17%	98.25%	98.00%	98.25%





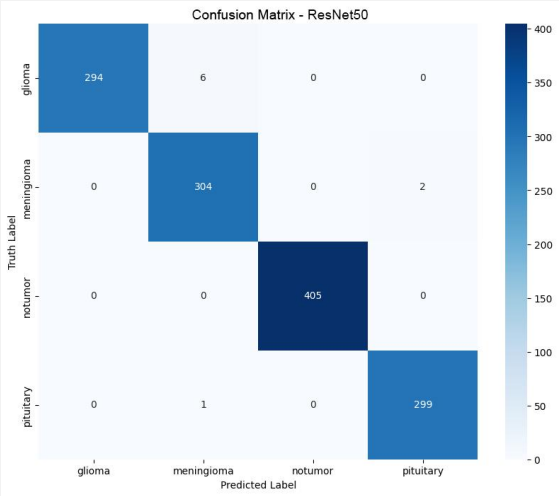


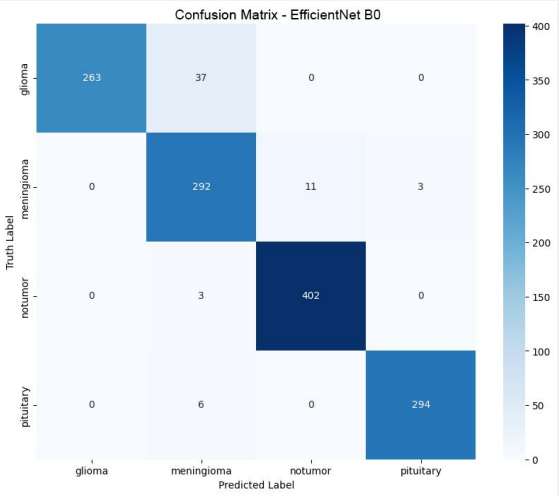


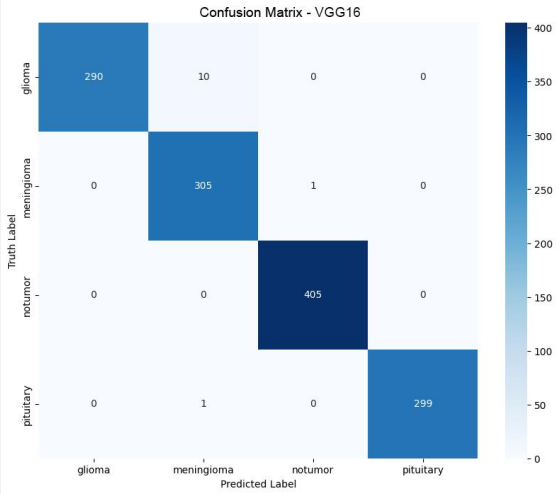


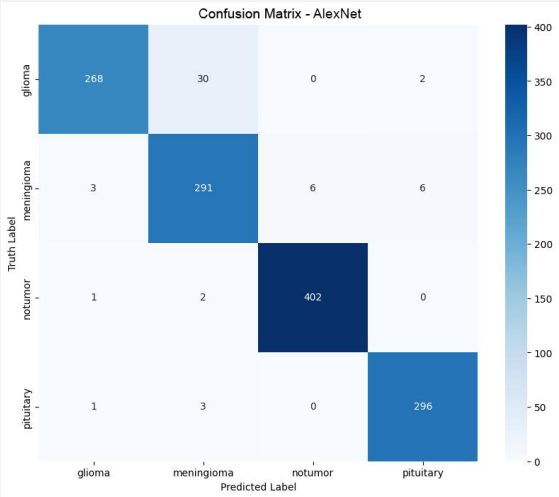
CNN Models Performance Analysis

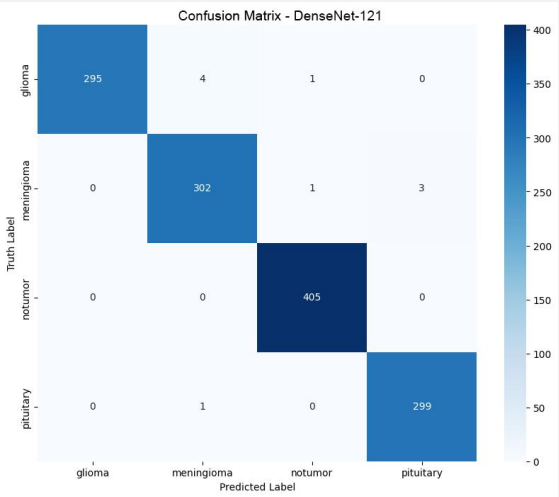
Model Name	Accuracy	Precision	Recall	F1-Score
ResNet50	99.31%	99.25%	99.25%	99.50%
EfficientNet	95.42%	95.50%	95.00%	95.00%
VGG16	99.08%	99.25%	99.25%	99.00%
AlexNet	95.88%	95.75%	95.50%	95.75%
DenseNet	99.24%	99.25%	99.25%	99.25%





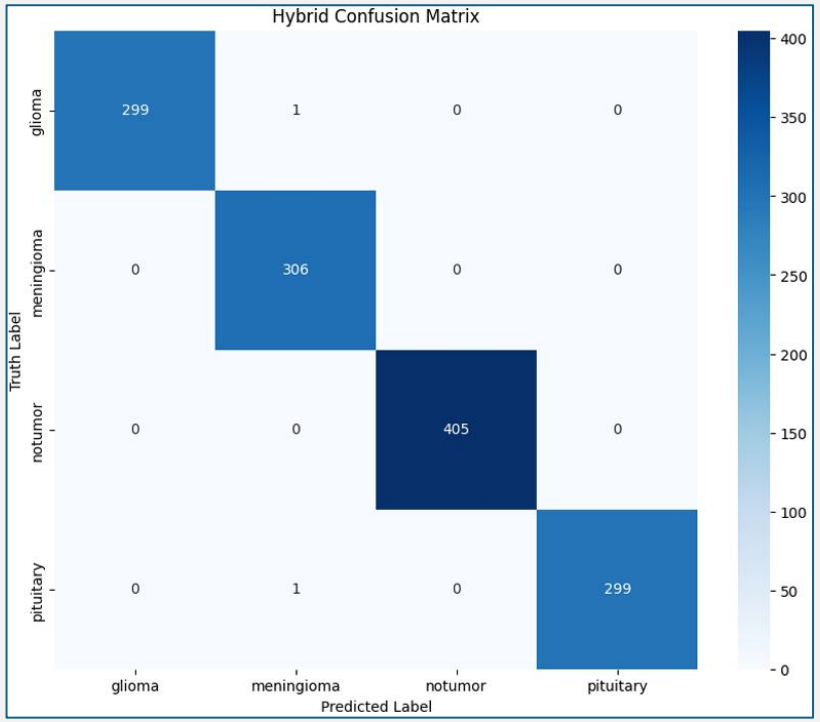




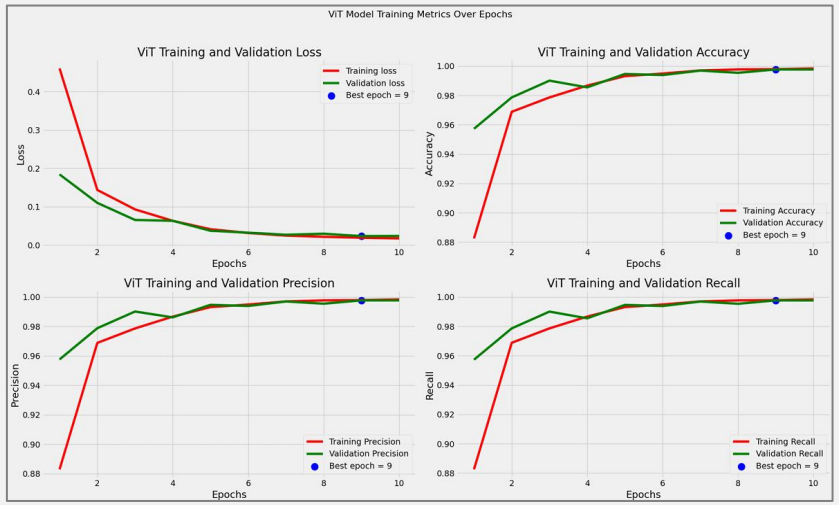


Hybrid Model Performance Analysis

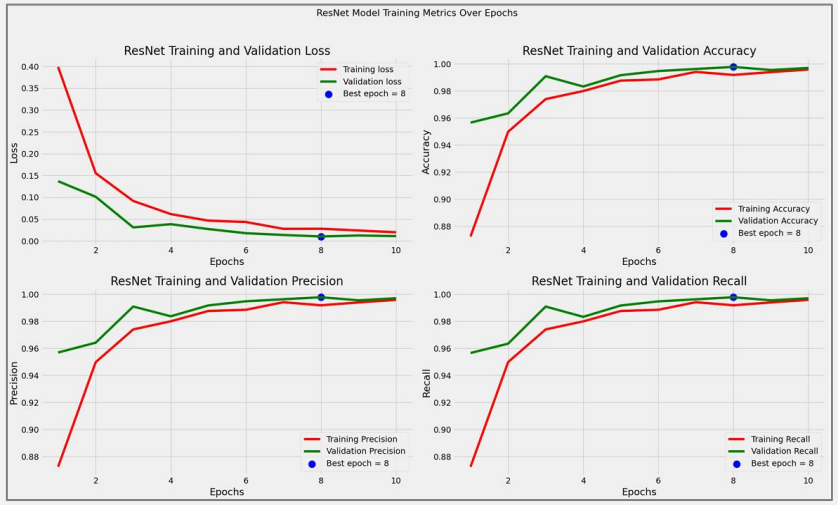
Model Name	Accuracy	Precision	Recall	F1-Score
VisionNet	99.85%	99.75%	100%	100%



ViT

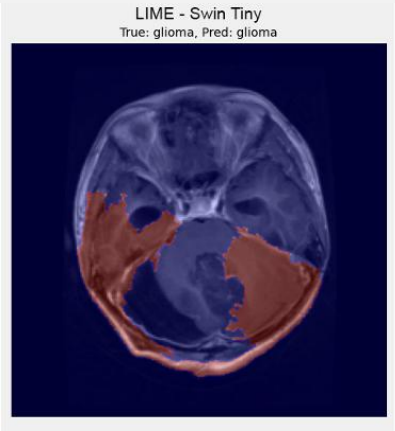


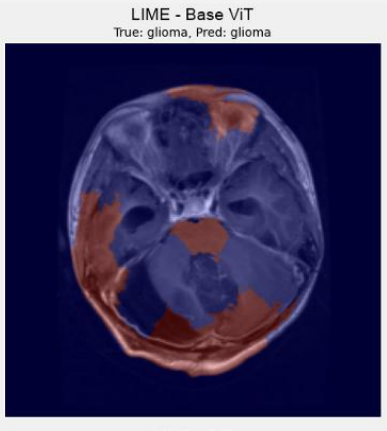
RestNet50

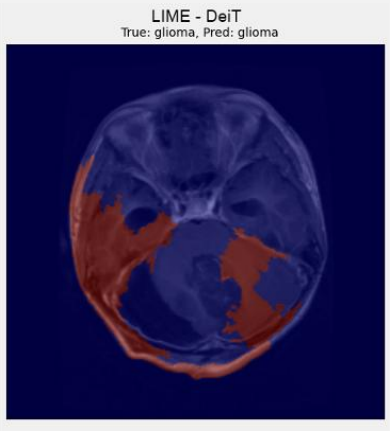


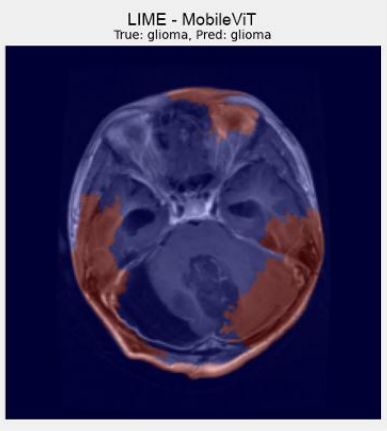
XAI- For Transparency And Validation

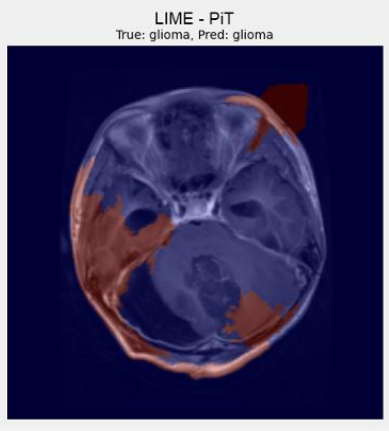
ViT Models

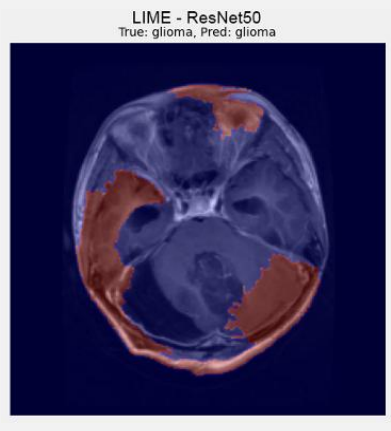


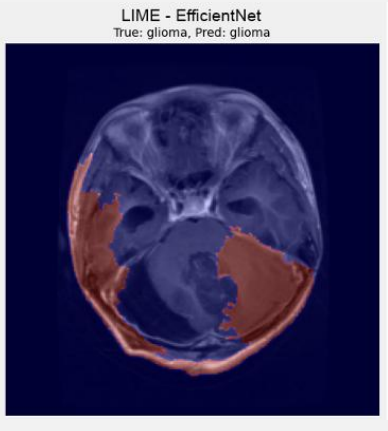


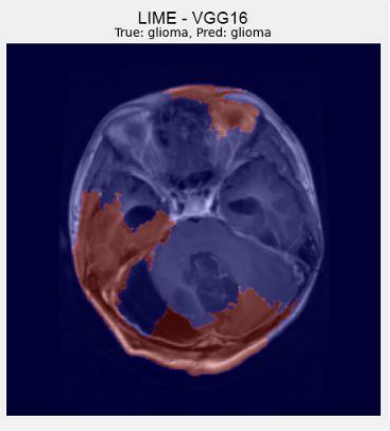


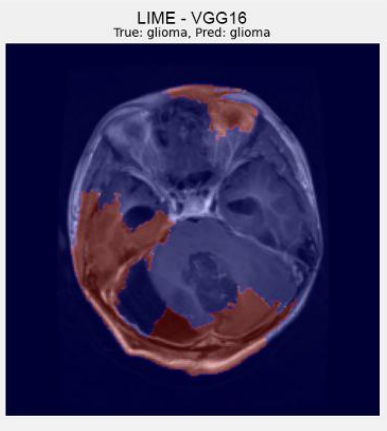


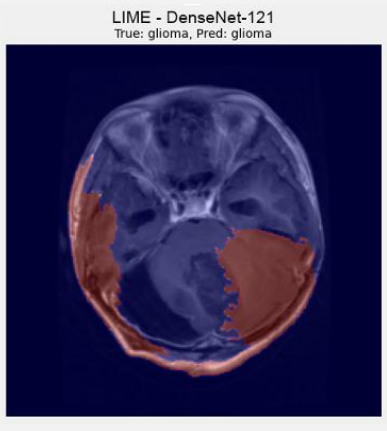




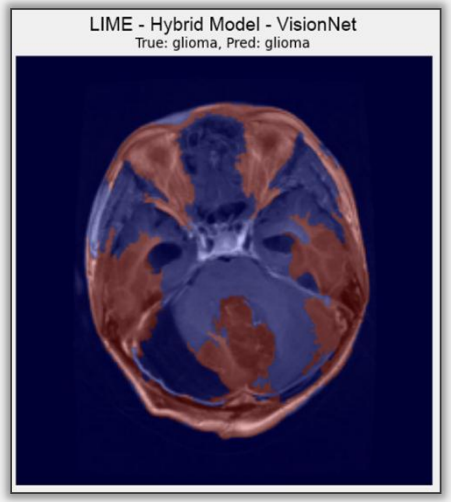








CNN Models



Hybrid Model

Future Work

AI Model Improvements

Advanced Explainability

Integrate SHAP and Grad-CAM hybrids for advanced explainable AI.

Multimodal Data

Extend the model to handle different types of data like CT and MRI scans.

Real-Time Deployment

Deploy the model on edge devices for real-time processing, focusing on architectures like MobileViT.

Self-Supervised Learning

Use self-supervised pretraining to improve performance when data is limited.

Conclusion

Key Achievement 1: Demonstrated ViTs' superiority over CNNs in accuracy (up to 99.77%) and global context capture, validated by LIME showing broader spatial reliance for tumor detection.

Key Achievement 2: Proposed and achieved top performance with ViT-ResNet50 hybrid at 99.85% accuracy and 100% F1-score, fusing global and local features for robust classification.

Key Achievement 3: Real-ESRGAN preprocessing enhanced all models' F1-scores by 1-2%, preserving anatomical details without resolution distortion—critical for clinical MRI quality.

Key Achievement 4: LIME insights confirm interpretable decisions (e.g., tumor borders over noise), bridging AI opacity to medical trust and enabling faster, reliable diagnoses.

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Thank You

Q & A